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Supplemental information

A cancer-associated fibroblast subtypes-based signature enables the evaluation of immunotherapy response and prognosis in bladder cancer

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Supplementary Information

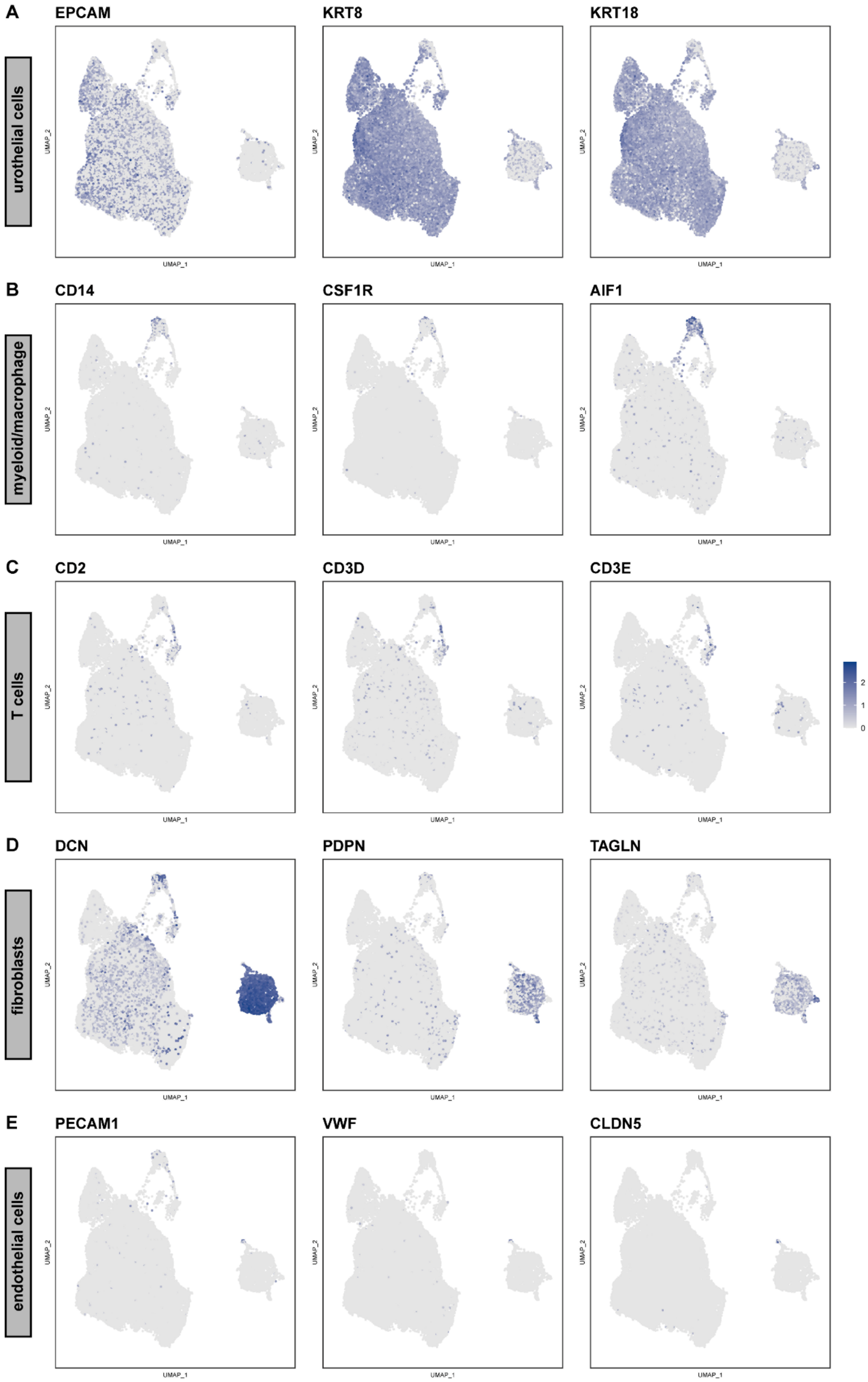

Figure S1. Canonical cell markers were used to identify the clusters in the single - cell

training dataset GSE135337, related to Figure 1.

(A) UMAP plots (gray to blue) for the expression of indicated markers in urothelial cells. (B) UMAP plots (gray to blue) for the expression of indicated markers in myeloid/macrophage cells. (C) UMAP plots (gray to blue) for the expression of indicated markers in T cells. (D) UMAP plots (gray to blue) for the expression of indicated markers in fibroblasts. (E) UMAP plots (gray to blue) for the expression of indicated markers in endothelial cells.

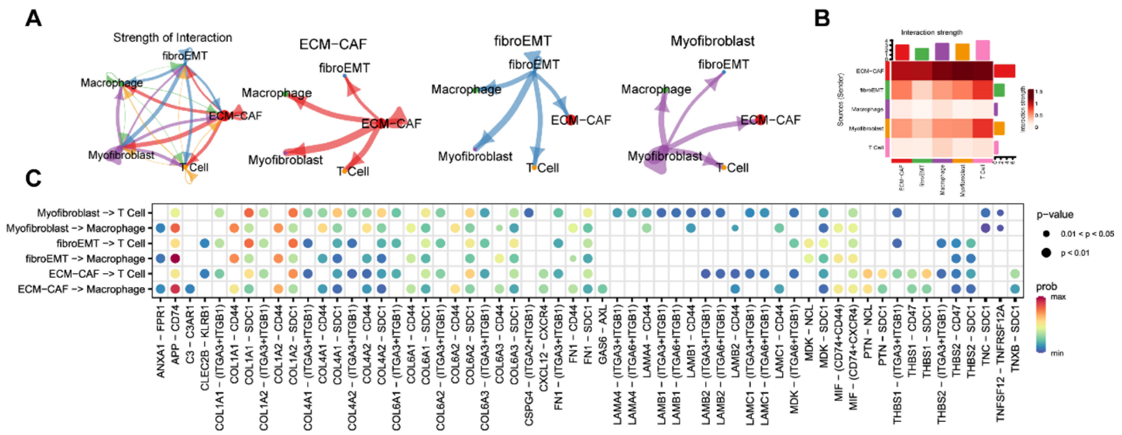

Figure S2. Cell-cell crosstalk in BLCA, related to Figure 1.

(A) Circos plot showing the cellular interactions between CAF subtypes and immune cells. (B) The heatmap showing the interaction strength between CAF subtypes and immune cells. (C) Dot plot showing enriched receptor-ligand pairs in CAF subtypes and immune cells.

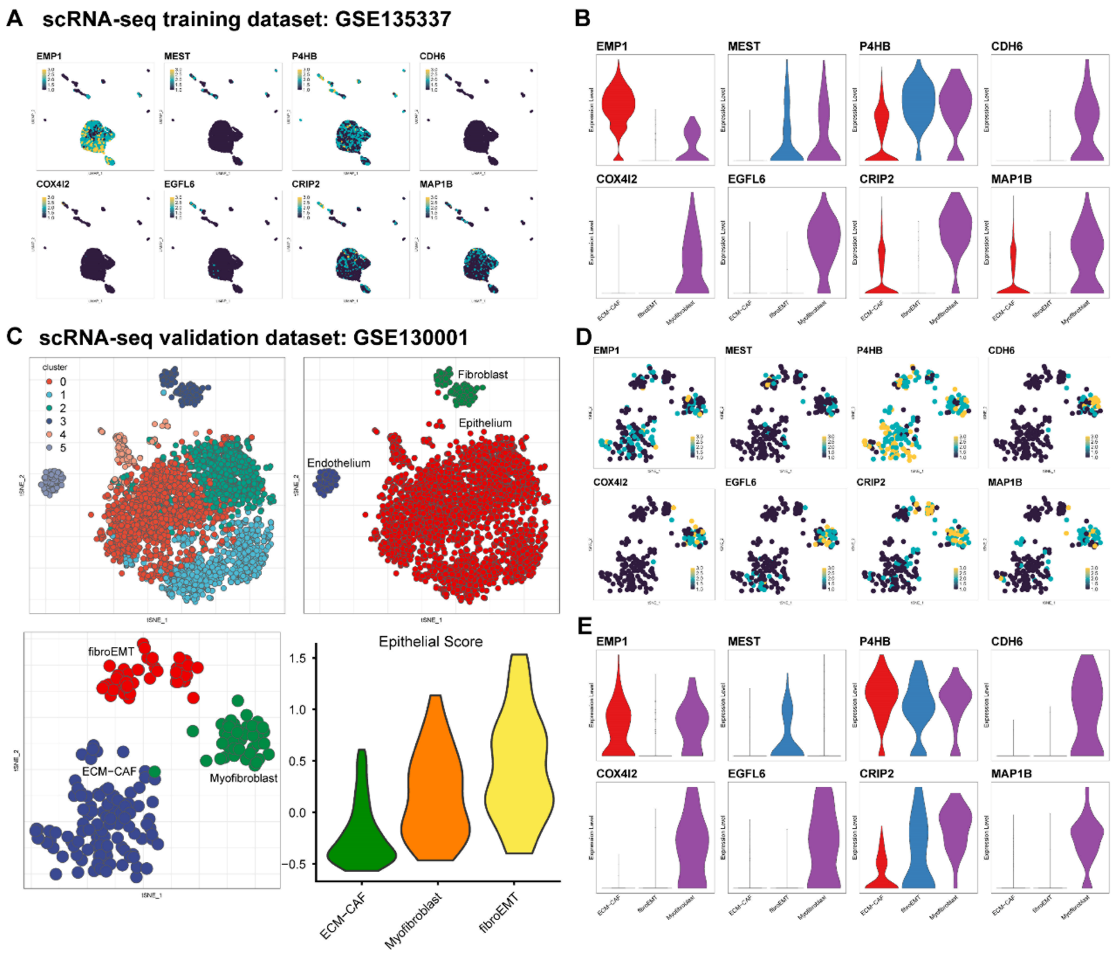

Figure S3. Identification the expression of 8 CAFs markers in scRNA-seq dataset, related to Figure 2.

(A) The expression of CAF-related markers in fibroblast subtypes of single-cell training set GSE135337. (B) The violin plots of CAF-related marker genes in single-cell training set GSE135337. (C) Reclustering of fibroblasts in scRNA-seq validation dataset GSE130001 represented as t-SNE plot. (D) t-SNE plots of selected 8 CAF-related markers in scRNA-seq GSE130001 dataset. (E) Violin plots of selected CAF-related markers in scRNA-seq GSE130001 dataset.

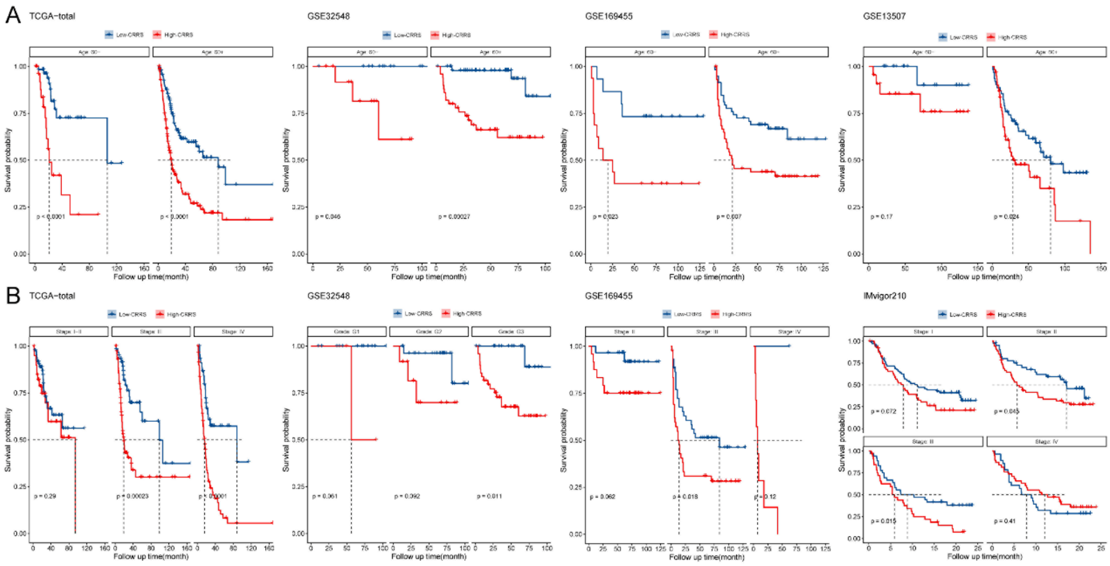

Figure S4. Subgroup survival analysis in high- and low-CRRS groups, related to Figure 2.

(A) The survival analysis of high- and low-CRRS groups with age < 60 or age ≥ 60 in TCGA-BLCA (n=393), GSE32548 (n=127), GSE169455 (n=148) and GSE13507 (n=165) cohorts (Log-rank test). (B) The survival analysis of high- and low-CRRS groups with different stages in TCGA-BLCA (n=393), GSE32548 (n=127), GSE169455 (n=148) and IMvigor210 (n=348) cohorts (Log-rank test).

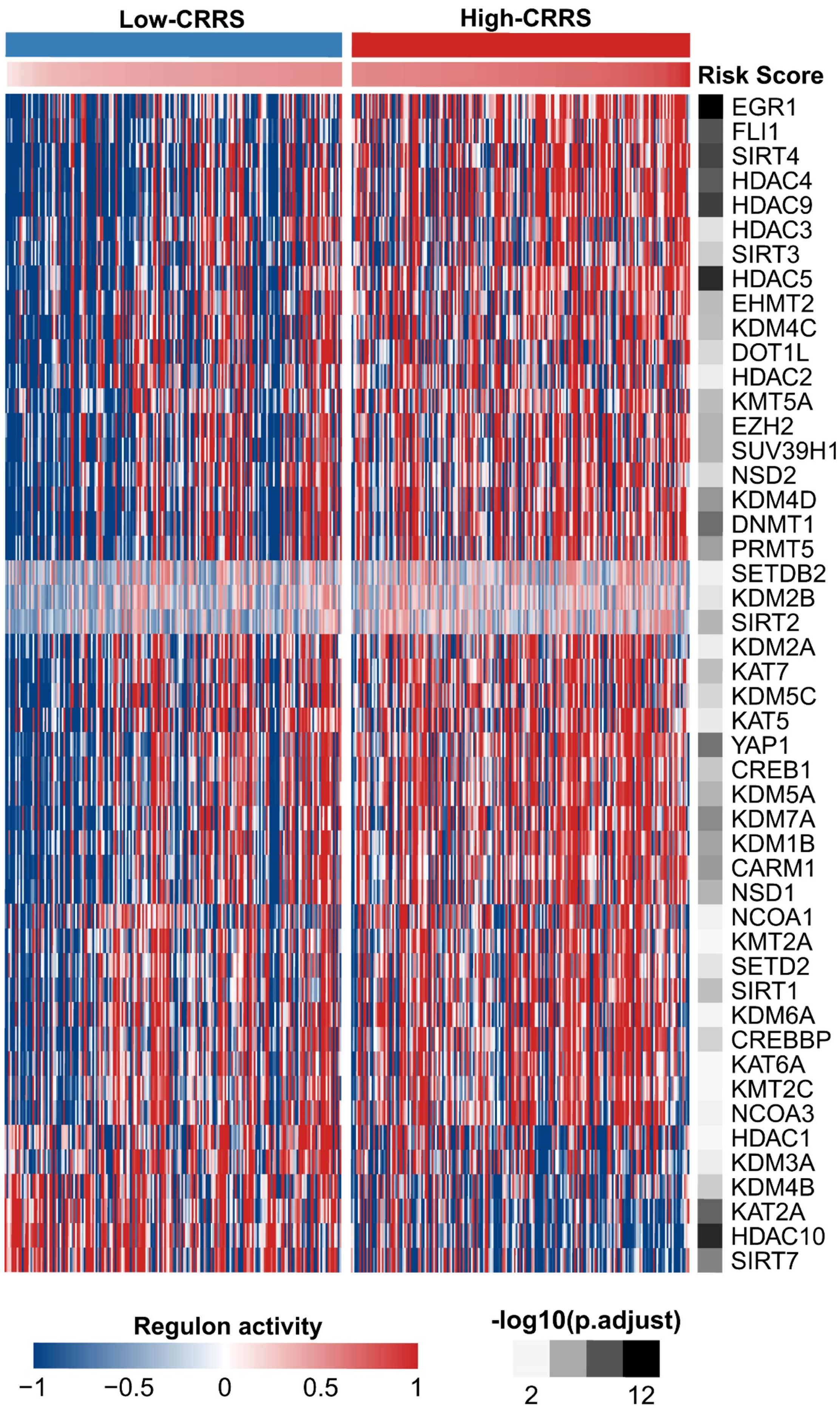
Low-CRRS
High-CRRS
Risk Score
EGR1
FLI1
SIRT4
HDAC4
HDAC9
HDAC3
SIRT3
HDAC5
EHMT2
KDM4C
DOT1L
HDAC2
KMT5A
EZH2
SUV39H1
NSD2
KDM4D
DNMT1
PRMT5
SETDB2
KDM2B
SIRT2
KDM2A
KAT7
KDM5C
KAT5
YAP1
CREB1
KDM5A
KDM7A
KDM1B
CARM1
NSD1
NCOA1
KMT2A
SETD2
SIRT1
KDM6A
CREBBP
KAT6A
KMT2C
NCOA3
HDAC1
KDM3A
KDM4B
KAT2A
HDAC10
SIRT7
Regulon activity
−1
−0.5
0
0.5
1
-log10(p.adjust)
2
12

Figure S5. The heatmap of potential regulators associated with chromatin remodeling in high- and low-CRRS groups (n=393, Wilcoxon rank sum test, FDR-adjustment), **related to Figure 4.**

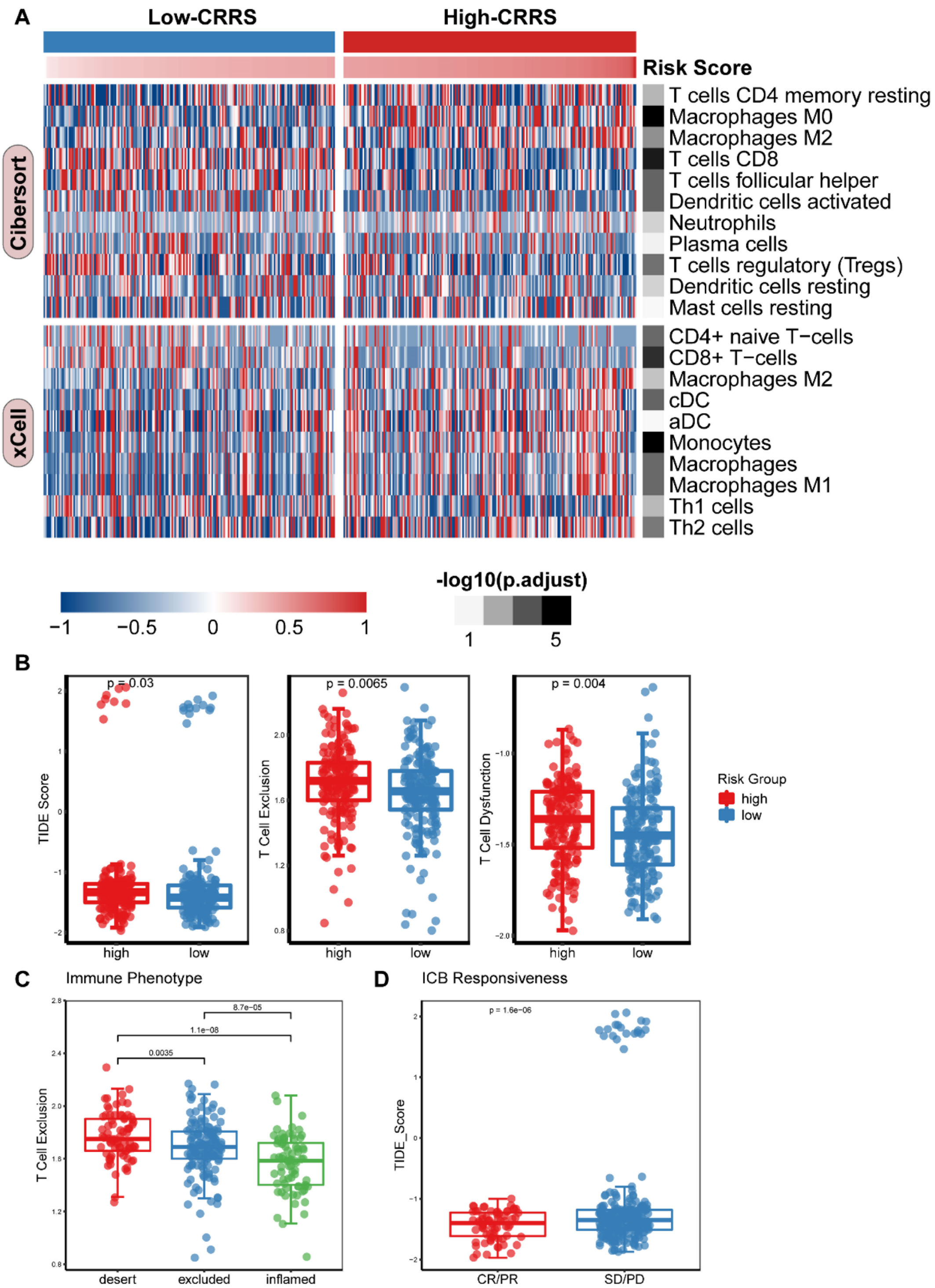

Figure S6. CRRS related to the immune phenotype and ICB response, related to Figure

6.

(A) The heatmap of the correlation between CRRS and immune infiltration in TCGA cohort (n=393, Wilcoxon rank sum test, FDR-adjustment). (B) The correlation between CRRS and TIDE score, T Cell Exclusion, T Cell Dysfunction in IMvigor210 cohort (n=348, Wilcoxon rank sum test). (C) The correlation between immune phenotype and T Cell Exclusion score in IMvigor210 cohort (n=348, Nemenyi test). (D) TIDE score in different ICB response groups in IMvigor210 cohort (n=348, Wilcoxon rank sum test).

Table S6. Gene sets used in this study**, related to Figure 1, Figure 4 & Figure 5.**

Term	**Gene list**
Early cell cycle	CCND1, CCND2, CCND3, RBL2, ID1, ID2, ID3, WEE1
Late cell cycle	CDK1, CDK4, CDK2, CCNE1, CDC20, CCNB2, CCNB1, CCNA2, BUB1, CDC25A, CCNE2, MYBL2, FOXM1, PLK1
Uroplakins	UPK1B, UPK3A, UPK1A, UPK2, UPK3B
Cancer stem cell markers	PROM1, ALDH2, ALDH1A1, ALDH1A3, CD47, ALDH1A2, NES, THY1, RPSA, SHH, ITGA6, CD44
partial EMT (p-EMT)	SERPINE1, TGFBI, MMP10, LAMC2, P4HA2, PDPN, ITGA5, LAMA3, CDH13, TNC, MMP2, EMP3, INHBA, LAMB3, VIM, SEMA3C, PRKCDBP, ANXA5, DHRS7, ITGB1, ACTN1, CXCR7, ITGB6, IGFBP7, THBS1, PTHLH, TNFRSF6B, PDLIM7, CAV1, DKK3, COL17A1, LTBP1, COL5A2, COL1A1, FHL2, TIMP3, PLAU, LGALS1, PSMD2, CD63, HERPUD1, TPM1, SLC39A14, C1S, MMP1, EXT2, COL4A2, PRSS23, SLC7A8, SLC31A2, ARPC1B, APP, MFAP2, MPZL1, DFNA5, MT2A, MAGED2, ITGA6, FSTL1, TNFRSF12A, IL32, COPB2, PTK7, OCIAD2, TAX1BP3, SEC13, SERPINH1, TPM4, MYH9, ANXA8L1, PLOD2, GALNT2, LEPREL1,

	MAGED1, SLC38A5, FSTL3, CD99, F3, PSAP, NMRK1, FKBP9, DSG2, ECM1, HTRA1, SERINC1, CALU, TPST1, PLOD3, IGFBP3, FRMD6, CXCL14, SERPINE2, RABAC1, TMED9, NAGK, BMP1, ESYT1, STON2, TAGLN, GJA1
Differentiation	TMEM163, GATA3, BAMBI, PPARG, GRHL3, SNX31, DHRS2, HPGD, SPINK1, FOXA1, FOXQ1, BHMT, SCNN1G
FGFR3-coexpressed genes	C3orf54, CAPNS2, SEMA4B, WNT7B, DUOXA1, C16orf74, ZNF385A, SMAD3, SLC2A9, D4S234E, TP63, CLCA4, IRS1, SYTL1, PLCH2, SSH3, FGFR3, PTPN13, DUOX1, TMPRSS4
EGFR ligands	EGFR, AREG, AREGB, EREG, HBEGF, TGFA
Hypoxia	CAV1, COL5A1, ITGA5, P4HA2, SLC16A1, TGFBI, DPYSL2, SRPX, TRAM2, SYDE1, LRP1, PDLIM2, SAV1, AHNAK2, CAD, CYP1B1, DAAM1, DSC2, SLC2A3, FUT11, GLG1, GULP1, LDLR, THBS4
DNA replication	DNA2, FEN1, LIG1, MCM2, MCM3, MCM4, MCM5, MCM6, MCM7, PCNA, POLA1, POLA2, POLD1, POLD2, POLD3, POLD4, POLE, POLE2, POLE3, POLE4, PRIM1, PRIM2, RFC1, RFC2, RFC3, RFC4, RFC5, RNASEH1, RNASEH2A, RNASEH2B, RNASEH2C, RPA1, RPA2, RPA3, RPA4, SSBP1
Pan-fibroblast TGF-β	ACTA2, ACTG2, ADAM12, ADAM19, CNN1, COL4A1, CTGF, CTPS1, FAM101B, FSTL3, HSPB1, IGFBP3, PXDC1, SEMA7A, SH3PXD2A, TAGLN, TGFBI, TNS1, TPM1
Transcription factors relevant to BLCA	AR, PGR, ESR1, ESR2, PPARG, RARA, RARB, RARG, RXRA, RXRG, ERBB2, ERBB3, FGFR1, FGFR3, EGFR, FOXA1, FOXM1, GATA3, GATA6, HIF1A, KLF4, STAT3, TP63

potential regulators associated with chromatin remodeling	EGR1, HNF4A, YAP1, FLI1, KDM5B, CREB1, IRF3, DNMT1, DNMT3A, DNMT3B, KAT5, KAT7, KAT6A, KAT6B, KAT2A, KAT2B, EP300, CREBBP, NCOA1, NCOA3, HDAC1, HDAC2, HDAC3, HDAC4, HDAC5, HDAC6, HDAC7, HDAC8, HDAC9, HDAC10, HDAC11, SIRT1, SIRT2, SIRT3, SIRT4, SIRT5, SIRT6, SIRT7, SUV39H1, EHMT2, SETDB2, KMT2A, KMT2B, KMT2C, KMT2D, KMT2E, SETD2, NSD1, SMYD2, SMYD3, NSD3, NSD2, DOT1L, KMT5A, EZH2, PRDM14, CARM1, PRMT5, PRMT6, KDM1A, KDM1B, KDM2A, KDM2B, KDM3A, KDM3B, KDM4A, KDM4B, KDM4C, KDM4D, KDM4E, KDM5A, KDM5C, KDM5D, KDM6A, KDM6B, KDM7A, KDM8, ARID1A
Epithelial mesenchymal transition (Hallmark)	ABI3BP, ACTA2, ADAM12, ANPEP, APLP1, AREG, BASP1, BDNF, BGN, BMP1, CADM1, CALD1, CALU, CAP2, CAPG, CCN1, CCN2, CD44, CD59, CDH11, CDH2, CDH6, COL11A1, COL12A1, COL16A1, COL1A1, COL1A2, COL3A1, COL4A1, COL4A2, COL5A1, COL5A2, COL5A3, COL6A2, COL6A3, COL7A1, COL8A2, COLGALT1, COMP, COPA, CRLF1, CTHRC1, CXCL1, CXCL12, CXCL6, CXCL8, DAB2, DCN, DKK1, DPYSL3, DST, ECM1, ECM2, EDIL3, EFEMP2, ELN, EMP3, ENO2, FAP, FAS, FBLN1, FBLN2, FBLN5, FBN1, FBN2, FERMT2, FGF2, FLNA, FMOD, FN1, FOXC2, FSTL1, FSTL3, FUCA1, FZD8, GADD45A, GADD45B, GAS1, GEM, GJA1, GLIPR1, GPC1, GPX7, GREM1, HTRA1, ID2, IGFBP2, IGFBP3, IGFBP4, IL15, IL32, IL6, INHBA, ITGA2, ITGA5, ITGAV, ITGB1, ITGB3, ITGB5, JUN, LAMA1, LAMA2, LAMA3, LAMC1, LAMC2, LGALS1, LOX, LOXL1, LOXL2, LRP1, LRRC15, LUM, MAGEE1, MATN2, MATN3, MCM7, MEST, MFAP5, MGP, MMP1, MMP14, MMP2, MMP3, MSX1, MXRA5, MYL9, MYLK, NID2, NNMT, NOTCH2, NT5E, NTM,

	OXTR, P3H1, PCOLCE, PCOLCE2, PDGFRB, PDLIM4, PFN2, PLAUR, PLOD1, PLOD2, PLOD3, PMEPA1, PMP22, POSTN, PPIB, PRRX1, PRSS2, PTHLH, PTX3, PVR, QSOX1, RGS4, RHOB, SAT1, SCG2, SDC1, SDC4, SERPINE1, SERPINE2, SERPINH1, SFRP1, SFRP4, SGCB, SGCD, SGCG, SLC6A8, SLIT2, SLIT3, SNAI2, SNTB1, SPARC, SPOCK1, SPP1, TAGLN, TFPI2, TGFB1, TGFBI, TGFBR3, TGM2, THBS1, THBS2, THY1, TIMP1, TIMP3, TNC, TNFAIP3, TNFRSF11B, TNFRSF12A, TPM1, TPM2, TPM4, VCAM1, VCAN, VEGFA, VEGFC, VIM, WIPF1, WNT5A